

OM BHATTARAI

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EDUCATION

Jalpaiguri Government Engineering College, West Bengal

Bachelor of Technology in Computer Science and Engineering (CSE)

May 2024 – Present

YGPA: 9.01

Sunshine School, Birpara, West Bengal

Higher Secondary (ISC) | Madhyamik (ICSE)

Jul 2021 – Feb 2023

Class XII : 91% | Class X : 89.68%

EXPERIENCE

Indian Institute of Technology, Guwahati

Research Intern

Jun 2026 – Present

Guwahati, Assam

- Engineered high-throughput, end-to-end infrastructure for **NLP pipelines** (NER, POS Tagging, Text Classification) utilizing **Transformers** and **PyTorch**, optimizing underlying data structures to improve tokenization efficiency.
- Architected scalable engineering solutions for NLP systems, executing performance-driven optimizations and debugging to ensure deterministic reproducibility and high-reliability production deployments.

Ministry of Education's Innovation Cell (AICTE)-Smart India Hackathon 2024

Innovation & Problem-Solving Internship Program

Nov 2024 – Dec 2024

Remote

Completed a 1-month pre-incubation training under the **Smart India Hackathon (SIH 2024)** organized by the Ministry of Education's Innovation Cell and AICTE.

Attended expert-led sessions covering **Web 5.0 Basics, Blockchain Fundamentals, Data Structures & Databases, UI/UX Design, Code Testing & Optimization, Cybersecurity Essentials, Cloud Management (AWS, Azure)**, and **IPR Awareness**.

Gained hands-on exposure to the innovation lifecycle — from **idea conceptualization** to **prototype development** and **pitch presentation**.

PROJECTS

campAlign: AI Marketing Campaign Automation Platform

React, Node.js, MongoDB, n8n, OpenAI API

- Deployed a comprehensive email marketing solution, enhancing campaign creation and preview capabilities, resulting in over **500** successful campaigns executed through the platform.
- Designed high-performance **REST APIs** with authentication (**JWT**) ensuring secure synchronization of event-driven automation workflow triggers.
- Developed event-driven automation workflows using **n8n** to handle campaign generation, audience segmentation, and email delivery
- Implemented customer segmentation pipeline using **Python (Scikit-Learn)** to process **5,000+** records and enable targeted messaging

Distributed Edge Computing Optimization Simulator

Python, NumPy, Algorithm Design, Dataclasses, Object-Oriented Design

- Engineered a highly scalable Mobile Edge Computing (MEC) simulation framework in **Python**, accurately modeling distributed task processing, network latency, and power constraints for high-volume workloads of up to **100,000 concurrent tasks**
- Architected from-scratch implementations of advanced metaheuristic algorithms—including **Particle Swarm Optimization (PSO)**, **Differential Evolution (DE)**, and a custom **Hybrid PSO+DE**—to dynamically optimize task-offloading policies
- Designed a rigorous, compute-normalized benchmarking pipeline utilizing **NumPy** for vectorized state evaluation, analyzing and optimizing for critical system metrics such as **p99 tail latency**, energy efficiency, and server overload

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL
Databases: MongoDB, MySQL
Backend: Node.js, Express.js, Ejs, REST APIs, JWT, Docker
Tools: Git, GitHub, Postman, VS Code, Jupyter

Core CS: Data Structures and Algorithms (DSA), OOPs
AI / ML: Scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, NLP, LLMs
Frontend: HTML, CSS, React.js, Next.js, Redux, Tailwind CSS

ACHIEVEMENTS

- * **Smart India Hackathon 2024 Finalist:** Achieved a top **2.4%** rank among **490K+** entries.
- * Solved **350+** Problems across LeetCode and various coding platforms.
- * Published **5 Research Papers** across Springer, IEEE and Cureus Journals.
- * **Presented a Research Paper at ICRICN 2025**, an IEEE international conference, highlighting research contributions in Mobile Edge Computing and optimization algorithms.